

03 PMF AND CDF (notes p063-p087)

1. The probability mass function, pmf (p064)

$p(x) = P(X = x) = P(\{s \text{ in } S : X(s) = x\})$
 "for every possible value x , the probability of observing it"

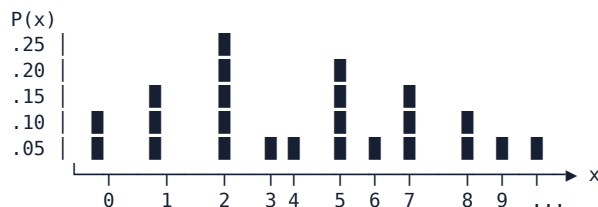
Two conditions (necessary AND sufficient):

- (i) $p(x) \geq 0$ every probability is non-negative
- (ii) $\sum p(x) = 1$ the probabilities over ALL possible x sum to 1

"The probability mass function of a discrete random variable describes the behaviour of the variable."

2. Worked: gas station pumps (p065)

x	0	1	2	3	4	5	6	
$p(x)$	.05	.10	.15	.25	.20	.15	.10	sum = 1.00 ✓



- (a) at most 2 $P(X \leq 2) = .05 + .10 + .15 = .30$
- (b) at least 3 $P(X \geq 3) = 1 - P(X \leq 2) = 1 - .30 = .70$
 or $.25 + .20 + .15 + .10 = .70$
- (c) between 2 and 5 inclusive $P(2 \leq X \leq 5) = .15 + .25 + .20 + .15 = .75$

THE cdf-arithmetic habit:

- "at most k " = $F(k)$ add up to k
- "at least k " = $1 - F(k-1)$ for an integer-valued X
- "between a, b " = $F(b) - F(a-1)$ for integer-valued X and integer a, b

3. Worked: the find-k table (p066) - the batch-1 assignment long Q2

x	0	1	2	3	4	5	6	7
$P(x)$	0	k	$2k$	$2k$	$3k$	k^2	$2k^2$	$7k^2 + k$

(i) FIND k

$$\begin{aligned} \sum p(x) &= k + 2k + 2k + 3k + k^2 + 2k^2 + 7k^2 + k \\ &= 9k + 10k^2 \\ 10k^2 + 9k &= 1 \rightarrow 10k^2 + 9k - 1 = 0 \\ 10k^2 + 10k - k - 1 &= 0 \rightarrow (k+1)(10k-1) = 0 \\ k &= 1/10 \quad (\text{or } -1, \text{ rejected: probability cannot be negative}) \end{aligned}$$

(ii) with $k = 1/10$:

$$\begin{aligned} P(X < 6) &= p(0) + p(1) + p(2) + p(3) + p(4) + p(5) \\ &= 0 + k + 2k + 2k + 3k + k^2 \\ &= 8k + k^2 \\ &= 0.80 + 0.01 = 0.81 \\ P(X \geq 6) &= 1 - 0.81 = 0.19 \\ P(0 < X < 5) &= p(1) + p(2) + p(3) + p(4) = k + 2k + 2k + 3k = 8k = 0.80 \end{aligned}$$

quick consistency check: $P(X < 6) + P(X \geq 6) = 0.81 + 0.19 = 1.00$ ✓

(iii) DISTRIBUTION FUNCTION: accumulate

$$F(x) = \begin{cases} 0 & x < 1 \\ 0.1 & 1 \leq x < 2 \end{cases}$$

0.3	$2 \leq x < 3$
0.5	$3 \leq x < 4$
0.8	$4 \leq x < 5$
0.81	$5 \leq x < 6$
0.83	$6 \leq x < 7$
1.0	$x \geq 7$

(iv) minimum a with $F(a) > 1/2$:

$$F(4) = 0.8 > 0.5 \rightarrow a = 4$$

($F(3) = 0.5$ is NOT > 0.5 , strict inequality)

WATCH THE $P(x)=0$ ROW

The batch-1 edition has $P(X=0)=0$, giving $10k^2+9k=1 \rightarrow k=1/10$.

The SECOND edition changed $P(X=0)$ to k , giving $12k+10k^2=1$

$\rightarrow k=0.0782$, but its own key still prints $k=1/10$ which does

NOT satisfy its own table. The second edition is broken

(errata 5.1). Study the batch-1 edition.

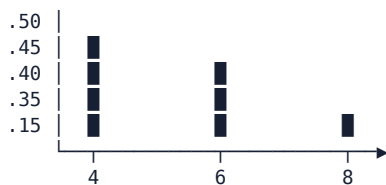
4. Worked: tune-ups, cylinders (p067-p069)

45% four-cylinder, 40% six-cylinder, 15% eight-cylinder

X = number of cylinders of the next car

x	:	4	6	8	
$p(x)$:		.45	.40	.15	sum 1.00 ✓

histogram picture:



$$P(X \geq 6) = .40 + .15 = .55$$

$$P(X > 6) = .15 \quad \leftarrow \text{the boundary moved}$$

5. Worked: building permits, proportional pmf (p070-p071)

Y = number of forms required, $p(y) = k y$, $y = 1, \dots, 5$

$$(a) \sum p(y) = k(1+2+3+4+5) = 15k = 1 \rightarrow k = 1/15$$

$$(b) P(Y \leq 3) = (1+2+3)/15 = 6/15 = 0.4$$

$$(c) P(2 \leq Y \leq 4) = (2+3+4)/15 = 9/15 = 0.6$$

$$(d) \text{ is } p(y) = y^2/50 \text{ a pmf? } \Sigma = (1+4+9+16+25)/50 = 55/50 \neq 1 \rightarrow \text{NO}$$

THE VALID-PMF TEST IS ALWAYS: sum over the support = 1 ?
If it does not sum to 1, it is not a pmf. Period.

6. Worked: flashlight batteries (p072-p073) - a pmf built from a process

90% of batteries acceptable (A), need 2 acceptable to work.

Y = number of batteries tested.

$$(a) p(2) = P(AA) = (.9)(.9) = .81$$

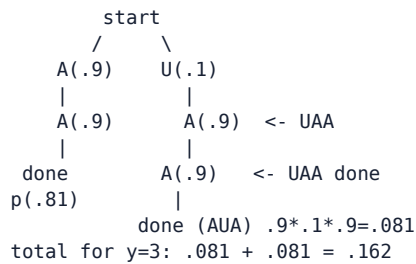
$$(b) p(3) = P(UAA \text{ or } AUA) = (.1)(.9)(.9) + (.9)(.1)(.9) = 2(.1)(.9)^2 = .162$$

$$(c) p(5): \text{ the 5th must be A, exactly one of the first four also A} \\ AUUUA, UAUUA, UUAUA, UUUAA = 4 \text{ outcomes}$$

$$p(5) = 4(.1)^3(.9)^2 = .00324$$

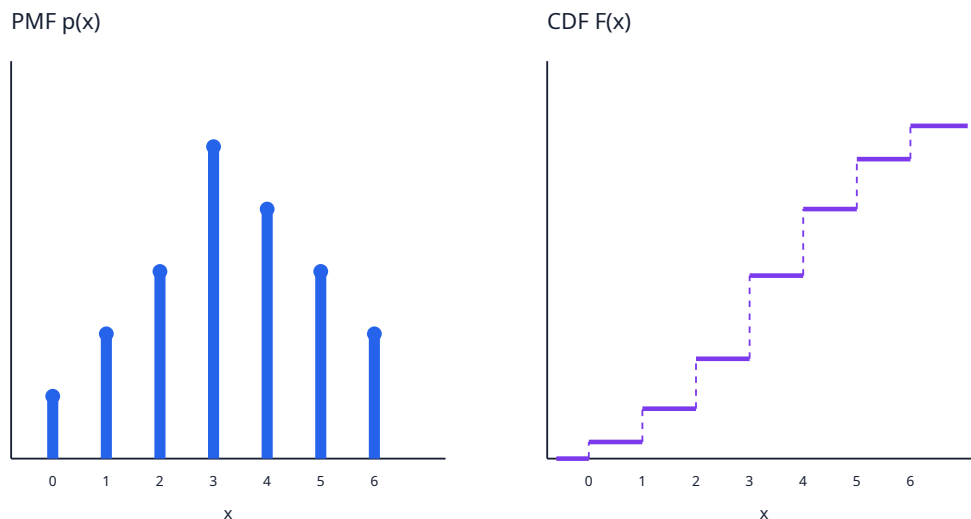
$$(d) \text{ general: } P(Y=y) = (y-1)(.1)^{y-2}(.9)^2, \quad y = 2, 3, 4, \dots$$

The tree for part (b):



7. The cumulative distribution function, cdf (p074-p078)

PMF accumulates into a CDF



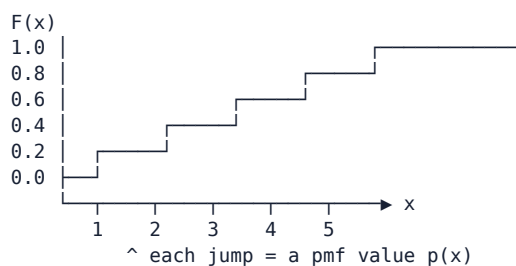
How a PMF accumulates into a CDF

$F(x) = P(X \leq x) = \sum_{\{y: y \leq x\}} p(y)$	(discrete)
$F(x) = P(X \leq x) = \int_{-\infty}^x f(y) dy$	(continuous)

PROPERTIES OF F

non-decreasing (it only accumulates)
 $0 \leq F(x) \leq 1$
 $F(-\infty) = 0, \quad F(+\infty) = 1$
 right-continuous

THE STEP-FUNCTION PICTURE (discrete cdf)



pmf AND cdf DETERMINE EACH OTHER

discrete: $p(x) = F(x) - F(x-)$ (the JUMP at x)

integer rv: $p(n) = F(n) - F(n-1)$

continuous: $f(x) = F'(x)$ (derivative)

8. THE key cdf theorem (p079)

for $a < b$:

$$P(a < X \leq b) = F(b) - F(a)$$

and the integer-valued version that catches people (p135):

$$P(a \leq X \leq b) = F(b) - F(a-1)$$

when X is integer-valued and a, b are integers

^^

the $a-1$ because $F(a-1)$ INCLUDES the value $a-1$, and

$P(a \leq X)$ must EXCLUDE everything below a

picture:

$F(b)$ = area/include everything up to and INCLUDING b

$F(a)$ = everything up to and including a <- but $P(a < X)$ excludes a

$F(a-1)$ = everything up to $a-1$ <- so this is the right one

...	a-1		a		...		b
drop this			keep			keep	

9. Worked: boards inspection (p080-p081)

Lots of 5 boards, 2 inspected. Boards 1,2 are the only defectives.

X = number of defective boards among the 2 inspected.

(a) all 10 selections: (1,2)(1,3)(1,4)(1,5)(2,3)(2,4)(2,5)(3,4)(3,5)(4,5)

(b) $P(X=0) = P\{(3,4)(3,5)(4,5)\} = 3/10 = .3$

$P(X=2) = P\{(1,2)\} = 1/10 = .1$

$P(X=1) = 1 - (.3 + .1) = .6$

(c) $F(0) = .30$

$F(1) = .90$

$F(2) = 1.00$

$F(x) = 0$	$x < 0$
.30	$0 \leq x < 1$
.90	$1 \leq x < 2$
1	$2 \leq x$

10. Worked: dice maximum M (p082-p083) - the elegant one

M = max of two dice

pmf: $p(1)=1/36, p(2)=3/36, p(3)=5/36, p(4)=7/36, p(5)=9/36, p(6)=11/36$

WHY the odd numbers? $M \leq m$ iff BOTH dice $\leq m \rightarrow m^2$ outcomes

$p(M = m) = [m^2 - (m-1)^2]/36 = (2m-1)/36$

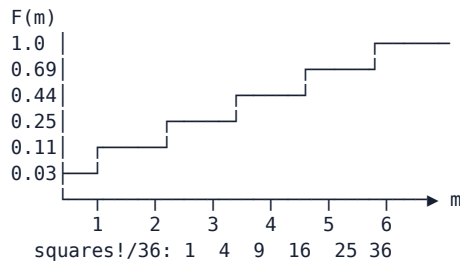
check $m=3$: $(2*3-1)/36 = 5/36 \checkmark$

cdf:

$F(m) = 0$	$m < 1$
1/36	$1 \leq m < 2$
4/36	$2 \leq m < 3$
9/36	$3 \leq m < 4$
16/36	$4 \leq m < 5$
25/36	$5 \leq m < 6$
1	$m \geq 6$

^ at integer support points $m=1, \dots, 6, F(m)=m^2/36$.

For real $1 \leq m < 6, F(m)=\text{floor}(m)^2/36$, as the steps above show.



11. Worked: cdf is non-decreasing (p085) - the proof question

SHOW: $x_1 < x_2 \rightarrow F(x_1) \leq F(x_2)$

proof:

$$\begin{aligned}
 F(x_2) &= P(X \leq x_2) \\
 &= P(\{X \leq x_1\} \cup \{x_1 < X \leq x_2\}) && \text{split the event} \\
 &= P(X \leq x_1) + P(x_1 < X \leq x_2) && \text{disjoint, so add} \\
 &\geq P(X \leq x_1) && \text{a probability is } \geq 0 \\
 &= F(x_1) && \checkmark
 \end{aligned}$$

AND: $F(x_1) = F(x_2)$ exactly when $P(x_1 < X \leq x_2) = 0$
 (the interval between them carries no probability)

pmf and cdf are done. Next: expectation and variance (notes p086-p111).